



QUANTITATIVE FINANCE

Internship subjects 2025 - 2026



ExiomPartners

Founded in 2019, Exiom Partners is a consulting firm offering its services to banks and insurance companies across all the sectors they encompass. Our values of excellence ensure the quality and sustainability of our services, while contributing to the development of our talents. Our innovative management model emphasizes trust and entrepreneurial spirit.

Based in Paris, the intern will join Exiom Partners' Quantitative team. He/she will work on one of the quantitative subjects described below. This is an end-of-study internship (5-6 months), and in addition to the internship subject, the intern will be assigned various tasks by the management to assess his/her technical, quantitative, and communication skills.

Given the nature of the subject, it is better suited for a quantitative profile meeting the requirements listed below:

- Strong quantitative skills (mathematics and finance)
- Good communication skills
- Fluency in English
- Good programming skills, a plus (object-oriented languages such as Python, C++)

If you are interested in joining a rapidly growing team and benefiting from personalized career support, do not hesitate to contact us at the following email address: recrutement.quant@exiompartners.com. Please indicate in the email the three subjects you are most interested in.

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1 High-Dimensional Correlation Matrices in Financial Data: Estimation, Regularization, and Applications

In quantitative finance, correlation matrices are fundamental tools for risk management, portfolio optimization, and factor modeling. They describe the dependence structure among asset returns and are central to methodologies such as Value-at-Risk computation, PCA-based factor models, and portfolio diversification analysis. However, in modern financial contexts—for example, when dealing with thousands of assets (equities, risk factors, or trading signals)—the estimation of a correlation matrix from limited historical data becomes problematic. A typical example is computing a $10,000 \times 10,000$ correlation matrix using only one year of daily data (250 observations): the sample covariance (or correlation) matrix is singular and contains substantial estimation noise. This "high-dimensional" regime, where the number of variables p is comparable to or exceeds the number of observations n , requires specialized mathematical tools to obtain meaningful estimators.

Objectives

The goal of this internship is to study, implement, and evaluate methods for robust estimation of correlation matrices in high-dimensional settings. Specifically, the intern will:

- Analyze the limitations of the empirical (sample) correlation matrix in the high-dimensional regime ($p \gg n$), including issues of instability, noise, and spurious correlations.
- Explore regularization techniques for correlation and covariance estimation, including, but not limited to:
 - Shrinkage estimators (e.g., Ledoit–Wolf, Oracle Approximating Shrinkage),
 - Factor-based and low-rank approximations (PCA, Random Matrix Theory denoising),
 - Sparse or structured estimators (Graphical Lasso, banding, tapering).
- Evaluate performance of these estimators on simulated and real financial data (e.g., equities, multi-asset portfolios, or factor exposures).
- Optionally, apply results to downstream financial problems (e.g., minimum-variance portfolios, risk budgeting, or stress testing).

Expected Work

- Theoretical study: Review of random matrix theory and high-dimensional statistics applied to covariance estimation,
- Numerical implementation: Development of estimators in Python,
- Empirical validation: Application on market data (Bloomberg, Refinitiv, FRED, Yahoo Finance),
- Deliverables: Technical report, code repository, and presentation comparing methods in terms of bias, stability, and impact on portfolio risk,

References

- [1] Ledoit, O., & Wolf, M. (2004). *A well-conditioned estimator for large-dimensional covariance matrices*. Journal of Multivariate Analysis, 88(2), 365–411.
- [2] Ledoit, O., & Wolf, M. (2020). *Analytical nonlinear shrinkage of large-dimensional covariance matrices*. Annals of Statistics, 48(5), 3043–3065.
- [3] Bun, J., Bouchaud, J.-P., & Potters, M. (2017). *Cleaning large correlation matrices: tools from random matrix theory*. Physics Reports, 666, 1–109.
- [4] Laloux, L., Cizeau, P., Bouchaud, J.-P., & Potters, M. (1999). *Noise dressing of financial correlation matrices*. Physical Review Letters, 83(7), 1467–1470.
- [5] Fan, J., Liao, Y., & Mincheva, M. (2013). *Large covariance estimation by thresholding principal orthogonal complements*. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 75(4), 603–680.
- [6] Bai, Z. D., & Silverstein, J. W. (2010). *Spectral Analysis of Large Dimensional Random Matrices*. Springer.
- [7] Bongiorno, C., Challet, D., & Loeper, G. (2021, 2023). *Cleaning the covariance matrix of strongly nonstationary systems with time-independent eigenvalues*, 2111-13109, <https://arxiv.org/abs/2111.13109>

2 Differential Machine Learning based CVA calculation

Banks involved in market and trading activities are exposed to counterparty credit risk and are required to actively quantify and manage this risk. In the framework of counterparty credit risk management, the CVA (*Credit Valuation Adjustment*) represents the market value of this risk. Traditionally, the calculation of CVA and its sensitivities with respect to market risk factors relies on stochastic methods, which are computationally expensive, notably via Monte Carlo simulations.

With the rise of artificial intelligence, financial institutions are exploring innovative approaches to obtain counterparty credit risk metrics quickly, accurately, and efficiently. In particular, **Differential Machine Learning** (an extension of supervised learning) can be implemented for CVA estimation and sensitivity calculation.

Objective

The main objective of this internship is to implement **Differential ML** to estimate CVA and compute its sensitivities for a derivatives portfolio. To achieve this goal, the topic is divided into three parts:

1. Understanding Differential Machine Learning and CVA calculation
2. Implementing Differential ML for the pricing of a derivative product
3. Estimating the CVA of a derivative product using Differential ML

1. Understanding Differential Machine Learning (DML) and CVA calculation

This is a documentation phase in which the intern must understand the foundations of DML and CVA calculation. The intern must also explore and understand AAD (Automatic Adjoint Differentiation), which is involved in the implementation of DML.

2. Implementing Differential ML for pricing a derivative product

Tackling CVA estimation using Differential Machine Learning, the intern will implement DML to evaluate a European call option. The principle is based on approximating the pricing function (depending on a set of input variables) with a neural network, as presented in (1), and the network is trained using labels obtained from sampling the payoff and the input variables.

3. Estimating the CVA of a derivative product using Differential ML

Considering a constant/known LGD (Loss Given Default) and assuming the default time τ has a density $d\Phi(0, t)$, it can be shown that the risk-neutral value of the CVA at $t = 0$ is:

$$\text{CVA} = \text{LGD} \times \int_0^T \mathbb{E}^{\mathbb{Q}}(D(0, t)V(t)^+ \mid \tau = t) d\Phi(0, t)$$

where $V(t)$ is the value of the cashflows at time t , and $D(0, t)$ is a discount factor.

By viewing CVA as a product, the idea is to identify its *payoff* and the differentials of the payoff with respect to input variables, and thereby implement Differential ML to estimate its fair value and compute sensitivities.

References

- [1] Brian Huge and Antoine Savine, *Differential Machine Learning*, 2020, <https://arxiv.org/abs/2005.02347>.

3 Bermudean pricing using Deep Learning

The pricing and valuation of exotic options are areas of focus for both investors and practitioners. Among these, Bermudan options offer the flexibility of early exercise at predetermined dates and therefore stand in the frontiers between European and American options.

As a quantitative researcher intern, you will have the opportunity to first explore the classic pricing methods of Bermudan options such as Longstaff–Schwartz and tree-based algorithms. These can, however, become computationally expensive in high-dimensional settings.

Recent advances in machine learning and deep learning provide promising alternatives. Neural networks and reinforcement learning can efficiently approximate the continuation value and optimal exercise strategy even in high-dimensional spaces.

This internship will explore these alternatives and contribute to our Quantitative Finance Python library by developing and benchmarking their architectures for pricing Bermudan options.

References

- [1] Longstaff, F. A., & Schwartz, E. S. (2001), *Valuing American Options by Simulation: A Simple Least-Squares Approach*. Review of Financial Studies.
- [2] Becker, S., Cheridito, P., & Jentzen, A. (2019), *Deep Optimal Stopping*. Journal of Machine Learning Research.
- [3] Becker, S., Cheridito, P., & Jentzen, A. (2020), *Pricing and Hedging American-Style Options with Deep Learning*. Applied Mathematical Finance.

4 Estimating Future Hedging Costs within the Prudent Valuation Framework

In an increasingly stringent regulatory environment, prudent valuation has become a critical concern for financial institutions. Prudent valuation requires a conservative approach to asset valuation, accounting for market uncertainties and the cost of managing those risks. A key aspect is the estimation of future costs associated with hedging adjustments, which are necessary to minimize long-term market risk. This involves optimal hedging strategies as well as the concept of replication—the construction of a portfolio that mirrors the payoff of a specific financial derivative, ensuring minimal risk exposure.

The article *Future Hedging Costs: Back to Basics* by Patry and Mozley (2021) presents a framework incorporating future hedging costs into the valuation process. It determines the trade-off between hedging costs and hedging errors due to discrete hedging for a carefully chosen utility function. This approach provides equations to solve for the expectation of the hedging costs and the standard deviation of the hedging errors, allowing the determination of the optimal risk management strategy without resorting to numerical techniques. This methodology could be utilized by regulators to define quantities relevant to the calculation of future hedging costs in the context of prudent valuation.

Internship Objectives

The objective of this internship is to study, model, and implement methods for estimating future hedging costs in the context of prudent valuation. Key tasks include:

1. **Theoretical Study:** Understand the principles and methodologies presented in *Future Hedging Costs: Back to Basics*, including the interaction between hedging costs, hedging errors, and replication strategies.
2. **Model Development:** Design a framework to calculate future hedging costs, incorporating market dynamics, regulatory constraints, and risk management principles.
3. **Replication Analysis:** Apply replication strategies to evaluate how effectively portfolios track derivative exposures, identifying potential gaps and inefficiencies.
4. **Simulation and Implementation:** Use market data-driven simulations to assess the impact of future hedging costs on prudent valuation metrics, ensuring robustness and practical applicability.
5. **Risk Management Recommendations:** Propose guidelines for integrating future hedging cost estimation into regulatory and internal risk management processes.

Outcome

By the end of the internship, the student will have developed a computational framework to estimate future hedging costs and assess their impact on prudent valuation. The project will provide insights into risk-adjusted pricing, optimal hedging, and regulatory compliance, equipping the intern with practical tools applicable in risk management and derivative pricing teams.

References

- [1] Patry, C., & Mozley, J. (2021), *Future Hedging Costs: Back to Basics*. SSRN.
- [2] Basel Committee on Banking Supervision (2019), *Prudent valuation guidance (CAP50)*. Bank for International Settlements.
- [3] European Banking Authority (2024), *Consultation Paper on Amendments to the RTS on Prudent Valuation*.
- [4] Albanese, C., Bénézet, C., & Crépéy, S. (2022), *Hedging Valuation Adjustment and Model Risk*.
- [5] Hull, J. (2022), *Options, Futures, and Other Derivatives*. Pearson.

5 Building Arbitrage-Free Implied Volatility Surface by Entropy Minimization

Local Volatility models and Local Stochastic Volatility models ideally require a full continuum in expiry and strike of arbitrage-consistent European option prices as input. In practice, we, of course, only observe a discrete set of option prices.

It is well-known that interpolation and extrapolation of a two-dimensional implied volatility surface is a non-trivial problem, particularly if one wishes to preserve characteristics that guarantee arbitrage-free option prices. Previous attempts to solve the problem include for instance parametric interpolation and extrapolation, where parametric data models such as [1] are fitted to observed implied volatilities and subsequently used for interpolation and extrapolation. But they only lead to an approximative fit and do not ensure that implied volatilities are within market bid/ask spreads.

In 2023, Hadrien De March and Pierre Henry-Labordère built in [2] an arbitrage-free implied volatility surface based on relative entropy minimization. They show that for any maturity of a surface, an arbitrage-free implied volatility slice can be deduced from a distribution, solving a strictly convex unconstrained minimization. The function to minimize depends on a set of European call price bounds, and an unspecified prior measure to be the positive Gaussian measure for easier calculation. This paper also proves numerical robustness of this minimization, which is crucial especially for industrial use, within an arbitrage-free volatility surfaces builder inside a pricing library. Finally, this method allows to fit market quotes perfectly within bid/ask spreads.

This internship would consist in the development of such a model within our Python library for quantitative finance. The article [3] may be studied to begin with to grasp the construction of arbitrage-free implied volatility surfaces. Further comparisons with other models will be possible, such as [1], which is already implemented in our Python library.

References

- [1] Jim Gatheral and Antoine Jacquier (March 17, 2013), *Arbitrage-free SVI volatility surfaces*.
- [2] Hadrien De March and Pierre Henry-Labordère (January 1, 2023), *Building arbitrage-free implied volatility: Sinkhorn's algorithm and variants*.
- [3] Fabien Le Floc'h (January 25, 2013), *Arbitrages in the volatility surface, interpolation and extrapolation*.

6 Pricing of CDOs

Collateralized Debt Obligations (CDOs) are derivative financial instruments created from portfolios of debt, which bundle and tranche the risks associated with the underlying credits. The pricing of these CDO tranches is complex, as it relies on modeling the correlation of credit defaults among the underlying assets, as well as quantifying the losses associated with different economic scenarios.

This internship proposes to explore various pricing approaches for CDOs, starting with structural credit models and copula models. The intern will begin by implementing a Gaussian copula, which is a market-standard model for estimating default correlations, and then investigate more advanced extensions, such as t-Student copulas and stochastic copulas, which better capture the heavy-tail behaviors observed during crises.

The intern will compare these approaches on market-quoted CDS Index tranches. This will involve calibrating these models on credit spread data, incorporating the implicit default correlation to adjust the prices of the various tranches. The project may also include an analysis of the sensitivity of the tranches to changes in correlations, which is crucial for understanding the propagation of credit risk within portfolios.

The intern will have the opportunity to code directly into Exiom Partners' internal Python pricing library, actively contributing to enhance the firm's ability to price credit derivatives. This internship will provide a unique opportunity to gain expertise in credit derivatives while developing practical skills in implementing complex stochastic models for pricing CDO tranches.

References

- [1] Li, D. X. (2000), *On default correlation: A copula function approach*. Journal of Fixed Income.
- [2] Duffie, D., & Gârleanu, N. (2001), *Risk and Valuation of Collateralized Debt Obligations*. Financial Analysts Journal, 57(1), 41–59.
- [3] Frey, R., & Backhaus, J. (2004), *Pricing and Hedging of Portfolio Credit Derivatives with Interacting Default Intensities*. Preprint, University of Leipzig.
- [4] Burtschell, X., Gregory, J., & Laurent, J.-P. (2008), *A comparative analysis of CDO pricing models*.

7 Option Pricing via Deep Reinforcement Learning with Path Signatures

Path-dependent options have long held a central place in trading and risk management activities of financial institutions. From a theoretical perspective, path dependence makes the problem intrinsically non-Markovian and thus more difficult to handle analytically. In practice, this difficulty is circumvented by increasing the dimension of the state system in order to restore a Markovian structure exploitable by Monte Carlo regression algorithms. While this approach allows the application of existing numerical tools, it often leads to a dimensional explosion and loss of computational efficiency.

In this context, path signatures offer a promising alternative: they allow the historical path of the underlying asset to be represented in a compact and rigorous way, while preserving essential temporal dependencies, thereby paving the way for more flexible valuation methods better suited to market data.

Objective

The main objective of this internship is to price path-dependent options using signatures and machine learning. The work will be divided into three axes:

1. Theoretical foundations: study of stochastic control problems for option valuation.
2. Literature review: theory of signatures and applications to financial problems.
3. Numerical applications: development of numerical methods using machine learning and signatures.

1. Theoretical Foundations

The intern will need to acquire a clear understanding of the theoretical formulation of the option pricing problem in a Markovian framework, mastering the analytical and probabilistic foundations of classical methods used for derivative pricing and hedging.

They will then analyze the theoretical limitations of these traditional approaches through a literature review, notably the work of *Chevalier, Ly Vath, & Mnif (2019)*, which highlights the constraints of Markovian frameworks for modeling path-dependent products or more realistic market dynamics.

2. Literature Review

The intern will need to gain a deep understanding of signature theory and its main applications. This phase will rely on foundational works, notably *Lyons, Caruana, & Lévy (2007)*, as well as surveys such as *Lyons (2014)* and *Chevyrev & Kormilitzin (2016)*.

The intern will also analyze the applications of signatures to financial problems through recent works, such as *Cuchiero, Gazzani, & Svaluto-Ferro (2023)*, and numerical methods for pricing exotic options (*Lyons, Nejad, & Perez Arribas, 2019*).

3. Numerical Applications

The final phase of the internship will consist of developing an original method based on the knowledge acquired. The work must be documented, and experiments implemented in Python following best practices for development and scientific reproducibility.

The intern will produce a methodological report detailing:

- Theoretical aspects developed,
- Design and functioning of the numerical method,
- Analysis of the results obtained.

References

- [1] Chevalier, E., Ly Vath, V., & Mnif, M. (2019), *Path-dependent American options*. Journal of Computational Finance.
- [2] Chevyrev, I., & Kormilitzin, A. (2016), *A Primer on the Signature Method in Machine Learning*.
- [3] Cuchiero, C., Gazzani, G., & Svaluto-Ferro, S. (2023), *Signature-based models: Theory and calibration*. SIAM.
- [4] Lyons, T. (2014), *Rough paths, signatures and the modelling of functions on streams*.
- [5] Lyons, T., Caruana, M., & Lévy, P. (2007), *Differential equations driven by rough paths*. Springer.
- [6] Lyons, T., Nejad, S., & Perez Arribas, I. (2019), *Numerical method for model-free pricing of exotic derivatives in discrete time using rough path signatures*. Taylor & Francis.

8 Gamma and Vega Hedging Using Deep Reinforcement Learning

Managing a portfolio of options, whether standard or exotic, requires decision-making regarding the hedging of various market risks associated with the trader's position. Generally, traders maintain a Delta-neutral portfolio at all times, whereas for Vega and Gamma, they have more flexibility.

The aim of this work is to propose a hedging strategy that accounts for transaction costs and the trader's risk aversion using Reinforcement Learning methods.

Objective

The main objective of this internship is to propose a hedging method that takes into account transaction costs and trader risk aversion based on Reinforcement Learning techniques. The work will be divided into three axes:

1. Literature review: Reinforcement Learning methods, option pricing, and calculation of Greeks.
2. Implementation: Simulation of the dynamic portfolio and implementation of the hedging strategy.
3. Results analysis: Influence of the cost function and pricing model, model robustness, etc.

1. Theoretical Foundations

Initially, the intern will need to acquire a clear understanding of the theoretical formulation of the option pricing problem within a Markovian framework. This stage aims to master the analytical and probabilistic foundations of classical methods used for derivative pricing and hedging.

The intern will then familiarize themselves with Reinforcement Learning methods, particularly DQN approaches that combine a neural network with quantile regression. Key references include:

- Jay Cao, Jacky Chen, Soroush Farghadani, John Hull, Zissis Poulos, Zeyu Wang, Jun Yuan, *Gamma and Vega Hedging Using Deep Distributional Reinforcement Learning*.
- Igor Halperin, *QLBS: Q-Learner in the Black-Scholes (-Merton) Worlds*.

2. Implementation: Dynamic Portfolio Simulation and Hedging Strategy

Next, the intern will develop a hedging method based on the knowledge acquired during the theoretical phase.

The intern will create a simulation tool for an options portfolio along with various pricing and sensitivity calculation tools, potentially leveraging the internal library. Subsequently, the intern will implement a Reinforcement Learning method based on: *Gamma and Vega Hedging Using Deep Distributional Reinforcement Learning* (Jay Cao et al.).

3. Numerical Applications

The final phase will focus on testing the implemented method (robustness, influence of model parameters, etc.). All work must be fully documented, and numerical experiments implemented in Python following best practices for development and scientific reproducibility.

The intern will produce a methodological report detailing:

- Theoretical aspects developed during the internship,
- Design and operation of the numerical method,
- Analysis of the results obtained.

References

- [1] Cao, J., Chen, J., Farghadani, S., Hull, J., Poulos, Z., Wang, Z., & Yuan, J. (2023), *Gamma and Vega Hedging Using Deep Distributional Reinforcement Learning*.
- [2] Halperin, I. (Year), *QLBS: Q-Learner in the Black-Scholes (-Merton) Worlds*.

9 Stochastic Initial Margin Modeling for Efficient EEPE Computation in Counterparty Credit Risk

The calculation of counterparty credit risk metrics, such as Expected Exposure (EE) and Expected Positive Exposure (EPE), is central to regulatory capital frameworks (e.g., Basel III/IV) and internal risk management practices. A key element in this computation is the netting of collateral, including Initial Margin (IM). IM is typically calculated based on risk-sensitive measures (e.g., Value-at-Risk or Conditional Value-at-Risk) and varies over time, introducing stochasticity into exposure profiles.

Incorporating stochastic initial margin into EEPE calculations can significantly improve the accuracy of risk assessment but is computationally demanding, as it requires simulating the joint dynamics of the portfolio value and margin calls over multiple time steps. To alleviate this computational burden, recent research has proposed approximation techniques such as Johnson's quantile estimation, where quantiles of the portfolio's distribution conditional on the current portfolio value V_t are approximated to efficiently compute the IM. Such approaches allow for faster Monte Carlo simulations while preserving accuracy in exposure metrics.

Internship Objectives

The internship aims to explore and implement advanced techniques for incorporating stochastic initial margin into EEPE calculations, focusing on both methodological rigor and computational efficiency. Specific objectives include:

1. **Literature Review:** Survey existing methods for stochastic IM modeling and quantile approximation, including Johnson's method and alternative techniques used in industry.
2. **Model Implementation:** Develop a framework for simulating portfolio exposure and stochastic IM, with a focus on derivative portfolios subject to bilateral margin agreements.
3. **Quantile Approximation Techniques:** Implement Johnson estimation of quantiles and compare its performance with full Monte Carlo calculations in terms of accuracy and computational cost.
4. **Performance Analysis:** Evaluate the impact of stochastic IM on EEPE profiles and risk measures. Assess the trade-off between computational efficiency and approximation error.
5. **Reporting and Recommendations:** Provide guidelines on the practical implementation of stochastic IM in EEPE calculations for financial institutions.

Outcome

By the end of the internship, the student will have developed a computational framework that integrates stochastic initial margin into EEPE calculations efficiently, providing insights into the impact of IM on counterparty risk metrics and offering practical tools for risk management applications.

References

- [1] McWalter, T., Kienitz, J., Nowaczyk, N., Rudd, R., & Acar, S. K. (2018), *Dynamic Initial Margin Estimation Based on Quantiles of Johnson Distributions*.
- [2] Albanese, C., & Andersen, L. (2014), *Counterparty Credit Risk, Collateral and Funding: With Pricing Cases for All Asset Classes*. Wiley.
- [3] Gregory, J. (2015), *The XVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital*. Wiley.
- [4] Brigo, D., & Mercurio, F. (2006), *Interest Rate Models – Theory and Practice*. Springer.
- [5] Burgard, C., & Kjaer, M. (2011), *Partial Differential Equation Representations of Derivatives with Bilateral Counterparty Risk and Funding Costs*. Journal of Credit Risk, 7(3), 1–19.
- [6] Pykhtin, M. (2010), *Stressed Expected Exposure: A Practical Guide to Monte Carlo Simulation and Approximation Techniques*. Risk Magazine.
- [7] Johnson, N. L., Kotz, S., & Balakrishnan, N. (1994), *Continuous Univariate Distributions, Vol. 1*. Wiley.

10 Pricing of CMS Spread Options

Constant Maturity Swap (CMS) spread options are interest rate derivatives that provide exposure to the spread between two swap rates of different maturities (for example, the 10-year and 2-year swap rates). These products are popular in structured interest rate products and are sensitive to the joint dynamics of the underlying swap rates, making their pricing both technically rich and challenging.

This internship proposes to explore and implement various pricing approaches for CMS spread options. The intern will begin by reviewing the fundamental properties of CMS rates and the convexity adjustments required to express them in a compatible measure for pricing. The first stage will consist of modeling the marginals with a stochastic volatility model such as the SABR model to better capture the volatility smile and skew observed in market quotes.

Once the marginals are modelled, the intern will start modelling the interest rate correlation, first with a Gaussian copula which captures the joint evolution of swap rates through correlated Brownian motions. The intern will then investigate more advanced frameworks. The calibration of these models will be performed using market data such as swaption volatilities, and the pricing will rely on numerical integration or Monte Carlo simulation techniques, depending on model complexity.

The intern will conduct a comparative analysis of the models' performance, focusing on accuracy, stability, and computational efficiency. Sensitivity analysis with respect to correlation and volatility parameters will also be carried out to assess their impact on option prices and risk management metrics.

The intern will have the opportunity to implement these models within Exiom Partners' internal Python pricing library, contributing to the enhancement of the firm's interest rate derivatives analytics. This internship offers a unique opportunity to deepen one's understanding of interest rate modeling, stochastic processes, and numerical pricing techniques applied to complex structured products.

References

- [1] Brigo, D., & Mercurio, F. (2006). *Interest Rate Models: Theory and Practice*. Springer.
- [2] Hagan, P. S., Kumar, D., Lesniewski, A. S., & Woodward, D. E. (2002). *Managing smile risk*. Wilmott Magazine.
- [3] Hagan, P. S. (2003). *Convexity conundrums: Pricing CMS swaps, caps, and floors*. Wilmott Magazine.

11 Pricing of Target Volatility Options

Target Volatility Options (TVO) are derivatives whose pay-off is adjusted to maintain a constant level of volatility exposure over the option's lifetime. By dynamically scaling the notional according to the realized volatility of the underlying asset, these options offer a mechanism to stabilize portfolio risk. This makes their valuation more intricate than that of standard European options, as the payoff depends jointly on both the underlying price and its realized volatility.

This internship aims to explore and implement different modeling and pricing approaches for TVOs. The intern will start by revisiting the theoretical foundations of stochastic volatility models, particularly the Heston model, as a natural framework to capture the dynamic relationship between the asset price and its instantaneous variance. The first objective will be to implement the pricing of Target Volatility Options under the Heston model using Monte Carlo simulations, where the volatility scaling is explicitly modeled along each simulated path.

Once the baseline model is established, we propose to focus on the pricing of Target Volatility Options within the framework of the lognormal fractional SABR model, a modern stochastic volatility model that captures both the smile effects and the long-memory behavior of volatility observed in financial markets. The project will begin with a theoretical review of the classical SABR model and its limitations in reproducing empirical volatility dynamics, followed by an introduction to its fractional extension, where the volatility process is driven by a fractional Brownian motion.

The intern will first implement Monte Carlo simulations of the lognormal fractional SABR model, carefully addressing numerical challenges related to the generation of correlated fractional Brownian increments and the discretization of the stochastic volatility process. Once the simulation framework is validated, the intern will use it to price Target Volatility Options, analyze their sensitivities, and compare results with the benchmark model.

The intern will compare the different methods in terms of accuracy, computational speed, and robustness, and perform sensitivity analyses with respect to volatility-of-volatility, correlation, and target volatility levels.

This internship provides a unique opportunity to work at the intersection of volatility modeling, numerical analysis, and equity derivatives pricing, offering valuable insights into both theory and practice. By tackling these analysis, the intern will actively contribute to enhancing Exiom Partners' ability to price interest rate derivatives and will have the opportunity to code directly into Exiom Partners' internal python pricing library.

References

- [1] Detlefsen, K., & Häusler, E. (2008). Pricing Target Volatility Options under Stochastic Volatility Models. *Quantitative Finance*.

- [2] Gatheral, J., Jaisson, T., & Rosenbaum, M. (2018). Volatility is Rough. *Quantitative Finance*, 18(6), 933–949.
- [3] Alos, E., Chatterjee, R., Tudor, S., & Wang, T.H. (2018). Target volatility option pricing in lognormal fractional SABR model.